



Cours MALG & MOVEX

Vérification mécanisée de contrats (III) (The ANSI/ISO C Specification Language (ACSL))

Dominique Méry
Telecom Nancy, Université de Lorraine

① Contracts

- Extending C programming language by contracts
- Playing with variables
- Ghost Variables
- Labels

② FIN MOVEX7

③ Generation of Verification Conditions

- WP calculus in Frama-c

④ FIN MALG1

- First annotation
- Second annotation

⑤ Memory Models in Frama-c

⑥ Logic Specification

⑦ Organisation of the verification process

⑧ Conclusion

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- requires
- assigns
- ensures
- decreases
- predicate
- logic
- lemma

① Contracts

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- ▶ The calling function should guarantee the required condition or precondition introduced by the clauses *requires* $P1 \wedge \dots \wedge Pn$ at the calling point.
- ▶ The called function returns results that are ensured by the clause *ensures* $E1 \wedge \dots \wedge Em$; ensures clause expresses a relationship between the initial values of variables and the final values.
- ▶ initial values of a variable v is denoted $\backslash old(v)$
- ▶ The variables which are not in the set $L1 \cup \dots \cup Lp$ are not modified.

Listing 1 – contrat

```
/*@ requires P1;...;requires Pn;  
   @ assigns L1;...;assigns Lm;  
   @ ensures E1;...;ensures Ep;  
   @*/
```

Examples of contract (1)

(Division)

Listing 2 – project-divers/annotation.c

```
/*@ requires x >= 0 && x <= 10;  
   @ assigns \nothing;  
   @ ensures x % 2 == 0 ==> 2*\result == x;  
   @ ensures x % 2 != 0 ==> 2*\result == x-1;  
*/  
int f(int x)  
{  
    int y;  
    y = x/2 ;  
    return(y);  
}
```

(Division)

Listing 3 – project-divers/annotationwp.c

```
/*@ requires 0 <= x && x <= 10;
   @ assigns \nothing;
   @ ensures x % 2 == 0 ==> 2*\result == x;
   @ ensures x % 2 != 0 ==> 2*\result == x-1;
   @*/

int f(int x)
{
  /*@ assert x % 2 == 0 ==> 2* (x / 2) == x; */
  /*@ assert x % 2 != 0 ==> 2* (x / 2) == x-1; */
  int y;
  /*@ assert x % 2 == 0 ==> 2* (x / 2) == x; */
  /*@ assert x % 2 != 0 ==> 2* (x / 2) == x-1; */
  y = x / 2;
  /*@ assert x % 2 == 0 ==> 2*y == x; */
  /*@ assert x % 2 != 0 ==> 2*y == x-1; */
  return(y);
  /*@ assert x % 2 == 0 ==> 2*y == x; */
  /*@ assert x % 2 != 0 ==> 2*y == x-1; */
}
```


Property to check

x is the value of the variable x when calling the function $x \geq$

$$0 \wedge x < 10; \Rightarrow \left(\begin{array}{l} x \% 2 = 0 \Rightarrow 2 \cdot (x/2) = x \\ x \% 2 \neq 0 \Rightarrow 2 \cdot (x/2) = x-1 \end{array} \right)$$

Proof Obligations from Frama-c

```
Goal Post-condition (file annotation.c, line 3) in 'annotation':
Let x_1 = x / 2.
Assume {
  Type: is_sint32(x) /\ is_sint32(x_1).
  (* Goal *)
  When: (x % 2) = 0.
  (* Pre-condition *)
  Have: (0 <= x) /\ (x <= 10).
}
Prove: (2 * x_1) = x.
Prover Alt-Ergo 2.6.0 returns Valid (Qed:0.71ms) (13ms) (29)
```

```
Goal Post-condition (file annotation.c, line 4) in 'annotation':
Let x_1 = x / 2.
Assume {
  Type: is_sint32(x) /\ is_sint32(x_1).
  (* Goal *)
  When: (x % 2) != 0.
  (* Pre-condition *)
  Have: (0 <= x) /\ (x <= 10).
}
Prove: (1 + (2 * x_1)) = x.
Prover Alt-Ergo 2.6.0 returns Valid (Qed:0.55ms) (14ms) (37)
}
```

A programme P *fulfils* a contract $(pre, post)$:

- ▶ P transforms a variable v à partir d'une valeur initiale v_0 and producing a final value v_f : $x_0 \xrightarrow{P} v_f$
- ▶ x_0 satisfait pre : $pre(v_0)$ and v_f satisfait $post$: $post(v_0, v_f)$
- ▶ $pre(v_0) \wedge v_0 \xrightarrow{P} v_f \Rightarrow post(v_0, v_f)$

```
requires  $pre(in_0)$   
ensures  $post(in_0, out_f)$   
variables  $in, out$   
 $type1 \quad f(type2 \ in_0)$   
local variables  $x$   
begin  
     $S(f)$   
return( $out$ );  
end
```

- ▶ $v = (in, out, x)$: in (input parameters), out (results), x (local variables)
- ▶ $pre(in_0) \wedge (in_0, out_0, v_0) \xrightarrow{P} (in_0, out_f, v_f) \Rightarrow post(in_0, out_f)$
- ▶ $pre(in_0) \wedge in = in_0 \wedge out = out_0 \Rightarrow [S(f)](post(in_0, out))$

Listing 4 – an1.c

```
//@ assert l1:  $P(x)$ ;  
  x = e(x);  
//@ assert l2:  $Q(x)$ ;
```

Listing 5 – an1.c

```
//@ assert l1: P(x);  
  x = e(x);  
//@ assert l2: Q(x);
```

► $P(x) \Rightarrow WP(x := e(x))(Q(x))$

Listing 6 – an1.c

```
//@ assert l1: P(x);  
  x = e(x);  
//@ assert l2: Q(x);
```

- ▶ $P(x) \Rightarrow WP(x := e(x))(Q(x))$
- ▶ $P(x) \Rightarrow Q[x \mapsto e(x)]$

Listing 7 – an1.c

```
//@ assert l1: P(x);  
  x = e(x);  
//@ assert l2: Q(x);
```

- ▶ $P(x) \Rightarrow WP(x := e(x))(Q(x))$
- ▶ $P(x) \Rightarrow Q[x \mapsto e(x)]$
- ▶ $P(x1) \Rightarrow Q[x \mapsto e(x1)]$ (renaming of free occurrences of x by $x1$)

Listing 9 – an1.c

```
//@ assert l1: P(x);  
  x = e(x);  
//@ assert l2: Q(x);
```

- ▶ $P(x) \Rightarrow WP(x := e(x))(Q(x))$
- ▶ $P(x) \Rightarrow Q[x \mapsto e(x)]$
- ▶ $P(x1) \Rightarrow Q[x \mapsto e(x1)]$ (renaming of free occurrences of x by $x1$)
- ▶ $P(x1) \wedge x = e(x1) \Rightarrow Q(x)$
- ▶ $P(x1) \wedge x = e(x1) \Rightarrow Q(x)$

Listing 10 – an1.c

```
//@ assert l1: P(x);  
  x = e(x);  
//@ assert l2: Q(x);
```

- ▶ $P(x) \Rightarrow WP(x := e(x))(Q(x))$
- ▶ $P(x) \Rightarrow Q[x \mapsto e(x)]$
- ▶ $P(x1) \Rightarrow Q[x \mapsto e(x1)]$ (renaming of free occurrences of x by $x1$)
- ▶ $P(x1) \wedge x = e(x1) \Rightarrow Q(x)$
- ▶ $P(x1) \wedge x = e(x1) \Rightarrow Q(x)$
- ▶ $\vdash P(x1) \wedge x = e(x1) \Rightarrow Q(x)$

Listing 11 – an1.c

```
//@ assert l1: P(x);  
  x = e(x);  
//@ assert l2: Q(x);
```

- ▶ $P(x) \Rightarrow WP(x := e(x))(Q(x))$
- ▶ $P(x) \Rightarrow Q[x \mapsto e(x)]$
- ▶ $P(x1) \Rightarrow Q[x \mapsto e(x1)]$ (renaming of free occurrences of x by $x1$)
- ▶ $P(x1) \wedge x = e(x1) \Rightarrow Q(x)$
- ▶ $P(x1) \wedge x = e(x1) \Rightarrow Q(x)$
- ▶ $\vdash P(x1) \wedge x = e(x1) \Rightarrow Q(x)$
- ▶ $P(x1) \wedge x = e(x1) \vdash Q(x)$

Listing 12 – an1.c

```
//@ assert l1: P(x);  
  x = e(x);  
//@ assert l2: Q(x);
```

- ▶ $P(x) \Rightarrow WP(x := e(x))(Q(x))$
- ▶ $P(x) \Rightarrow Q[x \mapsto e(x)]$
- ▶ $P(x1) \Rightarrow Q[x \mapsto e(x1)]$ (renaming of free occurrences of x by $x1$)
- ▶ $P(x1) \wedge x = e(x1) \Rightarrow Q(x)$
- ▶ $P(x1) \wedge x = e(x1) \Rightarrow Q(x)$
- ▶ $\vdash P(x1) \wedge x = e(x1) \Rightarrow Q(x)$
- ▶ $P(x1) \wedge x = e(x1) \vdash Q(x)$
 - Assume {
 - $P(x1)$
 - ▶ $x = e(x1)$
 - }
 - Prove: $Q(x)$


```
Goal Assertion
(file annotations14.c, line 5):
Assume {
  Type: is_sint32(x) /\ is_sint32(x-1).
  Have: x-1 = 1.
  Have: (1 + x-1) = x.
}
Prove: x = 2.
Prover Qed returns Valid
```

Verification conditions

```
Goal Assertion
(file annotations14.c, line 5):
Assume {
  Type: is_sint32(x) /\ is_sint32(x-1).
  Have: x-1 = 1.
  Have: (1 + x-1) = x.
}
Prove: x = 2.
Prover Qed returns Valid

void main(void){
  int x;
  x = 1;
  x = x + 1;
  /*@ assert x == 2; */
}
```

Verification conditions

```
Goal Assertion
(file annotations14.c, line 5):
Assume {
  Type: is_sint32(x) /\ is_sint32(x_1).
  Have: x_1 = 1.
  Have: (1 + x_1) = x.
}
Prove: x = 2.
Prover Qed returns Valid

void main(void){
  int x;
  x = 1;
  x = x + 1;
  /*@ assert x == 2; */
}
```

```
Goal Assertion
(file annotations15.c, line 7):
Assume {
  Type: is_sint32(x) /\ is_sint32(x_1)
      /\ is_sint32(x_2) /\ is_sint32(y).
  Have: x_2 = 1.
  Have: (1 + x_2) = x_1.
  Have: (2 * x_1) = y.
  Have: (x_1 + y) = x.
}
Prove: x = 6.
Prover Qed returns Valid
```


Verification conditions

```
Goal Assertion
(file annotations14.c, line 5):
Assume {
  Type: is_sint32(x) /\ is_sint32(x_1).
  Have: x_1 = 1.
  Have: (1 + x_1) = x.
}
Prove: x = 2.
Prover Qed returns Valid

void main(void){
  int x;
  x = 1;
  x = x + 1;
  /*@ assert x == 2; */
}
```

```
Goal Assertion
(file annotations15.c, line 7):
Assume {
  Type: is_sint32(x) /\ is_sint32(x_1)
      /\ is_sint32(x_2) /\ is_sint32(y).
  Have: x_2 = 1.
  Have: (1 + x_2) = x_1.
  Have: (2 * x_1) = y.
  Have: (x_1 + y) = x.
}
Prove: x = 6.
Prover Qed returns Valid

void main(void){
  int x,y;
  x = 1;
  x = x + 1;
  y = 2*x;
  x = x+y;
  /*@ assert x == 6; */
}
```

(Precondition)

Listing 13 – project-divers/annotation0.c

```
/*@ requires x >= 0 && x < 0;  
  @ assigns \nothing;  
  @ ensures \result == 0;  
  @*/  
int annotation0(int x)  
{  
    int y;  
    y = y / (x-x);  
    return(y);  
}
```

(Precondition)

Listing 14 – project-divers/annotation0wp.c

```
/*@ requires x >= 0 && x < 0;  
  @ assigns \nothing;  
  @ ensures \result == 0;  
  @*/  
int annotation(int x)  
{  
  /*@ assert y / (x-x) == 0; */  
  int y;  
  /*@ assert y / (x-x) == 0; */  
  y = y / (x-x);  
  /*@ assert y == 0; */  
  return(y);  
  /*@ assert y == 0; */  
}
```

Property to check

$$0 \leq x \wedge x \leq 10 \Rightarrow y/(x-x) = 0$$

Definition of a contract (specification)

- ▶ Define the mathematical function to compute (what to compute?)
- ▶ Define an inductive method for computing the mathematical function and using axioms.

(factorial what)

Listing 15 – project-factorial/factorial.h

```
#ifndef _A.H
#define _A.H
/*@ axiomatic mathfact {
    @ logic integer mathfact(integer n);
    @ axiom mathfact.1: mathfact(1) == 1;
    @ axiom mathfact_rec: \forall integer n; n > 1
    ==> mathfact(n) == n * mathfact(n-1);
    @ } */

/*@ requires n > 0;
    decreases n;
    ensures \result == mathfact(n);
    assigns \nothing;
*/
int codefact(int n);
#endif
```

Definition of a contract (programming)

- ▶ Define the program codefact for computing mathfact (How to compute?)
- ▶ Define the algorithm computing the function mathfact

(factorial how)

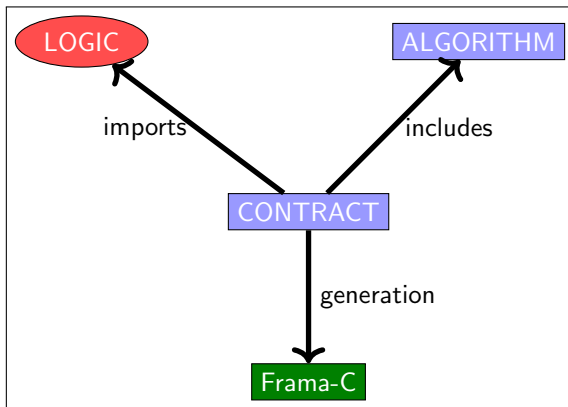
Listing 16 – project-factorial/factorial.c

```
#include "factorial.h"

int codefact(int n) {
    int y = 1;
    int x = n;
    /*@ loop invariant x >= 1 && x <= n && mathfact(n) == y * mathfact(x);
       loop assigns x, y;
       loop variant x;
    */
    while (x != 1) {
        y = y * x;
        x = x - 1;
    };
    return y;
}
```

Definition of a contract (approach)

- ▶ The specification of a function (mathfact) to compute requires to define it mathematically.
- ▶ The definition is stated in an axiomatic framework and is preferably inductive (mathfact) which is used in assertions or theorems or lemmas.
- ▶ The relationship between the computed value ($\backslash\text{result}$) and the mathematical value ($\text{mathfact}(n)$) is stated in the ensures clause :
$$\backslash\text{result} == \text{mathfact}(n)$$
- ▶ The main property to prove is $\text{codefact}(n) == \text{mathfact}(n)$: Calling codefact for n returns a value equal to $\text{mathfact}(n)$.



Listing 17 – contrat

```

/*@ requires P;
   @ behavior b1:
       @ assumes A1;
       @ requires R1 ;
       @ assigns L1;
       @ ensures E1;
   @ behavior b2:
       @ assumes A2;
       @ requires R2;
       @ assigns L2;
       @ ensures E2;
   @*/

```

(Pairs of integers)

Listing 18 – project-divers/structures.h

```
#ifndef _STRUCTURE_H
```

```
struct s {  
    int q;  
    int r;  
};
```

```
#endif
```

Division should not return silly expressions !

(Specification)

Listing 19 – project-divers/division.h

```
#ifndef _A.H
#define _A.H
#include "structures.h"
/*@ requires a >= 0 && b >= 0;
    @ requires b != 0;
    @ ensures 0 <= \result.r;
    @ ensures \result.r < b;
    @ ensures a == b * \result.q + \result.r;
*/
struct s division(int a, int b);
#endif
```

Division should not return silly expressions !

(Algorithm)

Listing 20 – project-divers/division.c

```
#include <stdio.h>
#include <stdlib.h>
#include "division.h"

struct s division(int a, int b)
{
    int rr = a;
    int qq = 0;
    struct s silly = {-1,-1};
    struct s resu;
    if (b == 0) {
        return silly;
    }
    else
    {
        /*@
        loop invariant
        ( a == b*qq + rr) &&
        rr >= 0;
        loop assigns rr,qq;
        loop variant rr;
        */
        while (rr >= b) { rr = rr - b; qq=qq+1;};
        resu.q = qq;
        resu.r = rr;
        return resu;
    }
}
```

Iteration Rule for PC

If $\{P \wedge B\}S\{P\}$, then $\{P\}\mathbf{while\ B\ do\ S\ od}\{P \wedge \neg B\}$.

- ▶ Prove $\{P \wedge B\}S\{P\}$ or $P \wedge B \Rightarrow \{S\}(P)$.
- ▶ By the iteration rule, we conclude that $\{P\}\mathbf{while\ B\ do\ S\ od}\{P \wedge \neg B\}$ without using WLP.
- ▶ Introduction of LOOP INVARIANTS in the notation.

Listing 21 – loop.c

```
/*@ loop invariant I1;  
    loop invariant I2;  
  
    ...  
    loop invariant In;  
    loop assigns X;  
    loop variant E;  
*/
```

(Invariant de boucle)

Listing 22 – project-divers/anno6.c

```
/*@ requires a >= 0 && b >= 0;  
    ensures 0 <= \result;  
    ensures \result < b;  
    ensures \exists integer k; a == k * b + \result;  
*/  
int rem(int a, int b) {  
    int r = a;  
    /*@  
        loop invariant  
        (\exists integer i; a == i * b + r) &&  
        r >= 0;  
        loop assigns r;  
    */  
    while (r >= b) { r = r - b; };  
    return r;  
}
```

- ▶ $\backslash old(x)$ is the value of the variable when the function is called.
- ▶ It can be used in the postcondition of the *ensures* clause.

(Modifying variables while calling)

Listing 23 – project-divers/old1.c

```
/*@ requires \valid(a) && \valid(b);  
   @ assigns *a,*b;  
   @ ensures  *a == \at(*a,Pre) +2;  
   @ ensures  *b == \at(*b,Pre)+\at(*a,Pre)+2;  
  
   @ ensures  \result == 0;  
*/  
int old(int *a, int *b) {  
    int x,y;  
    x = *a;  
    y = *b;  
    x=x+2;  
    y = y +x;  
  
    *a = x;  
    *b = y;  
    return 0 ;  
}
```

- ▶ $\backslash at(e, id)$ is the value of e at the control point id .
- ▶ id should occur before $\backslash at(e, id)$
- ▶ id is one of the possible expressions : Pre, Here, Old, Post, LoopEntry, LoopCurrent, Init
- ▶ $\backslash old(e)$ is equivalent to $\backslash at(e, Old)$

(label Pre)

Listing 24 – project-divers/at1.c

```
/*@
  requires \valid(a) && \valid(b);
  assigns *a,*b;
  ensures *a == \old(*a)+2;
  ensures *b == \old(*b)+\old(*a)+2;
*/
int at1(int *a, int *b) {
  //@ assert *a == \at(*a, Pre);
  *a = *a +1;
  //@ assert *a == \at(*a, Pre)+1;
  *a = *a +1;
  //@ assert *a == \at(*a, Pre)+2;
  *b = *b +*a;
  //@ assert *a == \at(*a, Pre)+2 && *b == \at(*b, Pre)+\at(*a, Pre)+2;
  return 0;
}
```


(autre label)

Listing 25 – project-divers/at2.c

```
void f (int n) {  
  for (int i = 0; i < n; i++) {  
    /*@ assert \at(i, LoopEntry) == 0; */  
    int j=0;  
    while (j++ < i) {  
      /*@ assert \at(j, LoopEntry) == 0; */  
      /*@ assert \at(j, LoopCurrent) + 1 == j; */  
    }  
  }  
}
```

(otherlabel)

Listing 26 – project-divers/change1.c

```
/*@ requires \valid(a) && *a >= 0;
   @ assigns *a;
   @ ensures *a == \old(*a)+2 && \result == 0;
*/
int change1(int *a)
{
    int x = *a;
    x = x + 2;
    *a = x;
    return 0;
}
```

- ▶ A variable called *ghost* allows to model a computed value useful for stating a model property : the ghost variable is hidden for the computer but not for the model.
- ▶ It should not change the semantics of others variables and should not change the effective variables.

(Bug)

Listing 27 – project-divers/ghost2.c

```
int f (int x, int y) {  
    //@ghost int z=x+y;  
    switch (x) {  
    case 0: return y;  
    //@ ghost case 1: z=y;  
    // above statement is correct.  
    //@ ghost case 2: { z++; break; }  
    // invalid, would bypass the non-ghost default  
    default: y++; }  
    return y; }  
  
int g(int x) { //@ ghost int z=x;  
    if (x>0){return x;}  
    //@ ghost else { z++; return x; }  
    // invalid, would bypass the non-ghost return  
    return x+1; }
```

(Ghost variable)

Listing 28 – project-divers/ghost1.c

```
/*@ requires a >= 0 && b >= 0;  
    ensures 0 <= \result;  
    ensures \result < b;  
    ensures \exists integer k; a == k * b + \result; */  
int rem(int a, int b) {  
    int r = a;  
    /*@ ghost    int q=0;    */  
    /*@  
        loop invariant  
        a == q * b + r &&  
        r >= 0 && r <= a;  
        loop assigns r;  
        loop assigns q;  
    // loop variant r;  
    */  
    while (r >= b) {  
        r = r - b;  
    /*@ ghost    q = q+1;    */  
    };  
    return r;  
}
```

In **Frama-C** and the **ACSL** language, the expression :

$$\backslash\text{at}(e, L)$$

evaluates an expression e in the memory state corresponding to a label L .

Main labels

- ▶ Pre : state at the beginning of the function
- ▶ Post : state at the end of the function
- ▶ Here : current state
- ▶ Old : initial value (often in ensures)
- ▶ LoopEntry : state before the loop
- ▶ LoopCurrent : start of the current iteration
- ▶ Init : initial state of the programme

Example with Old

```
void swap(int *a, int *b){  
    int tmp;  
  
    /*@  
        requires a != \null && b != \null;  
        assigns *a, *b;  
        ensures *a == \at(*b, Old);  
        ensures *b == \at(*a, Old);  
    */  
  
    tmp = *a;  
    *a = *b;  
    *b = tmp;  
}
```

In proofs with the **WP** plugin, we must compare :

- ▶ the state **before** execution
- ▶ the state **after** execution

Example :

$$a_{final} = b_{initial}$$

We therefore write :

```
*a == \at(*b,Pre)
```

This allows the prover to reason about the **initial values of the programme**.

Listing 29 – contrat

```
/*@ requires x >= 0;  
@ ensures \result == \old(x) ;  
@*/  
int f(int x)  
{  
    int y;  
    y=x;  
    x=x+1;  
    return(y);  
}
```

Listing 30 – contrat

```
/*@ requires x >= 0;  
@ ensures \result == \old(x) && \old(x) == x;  
@*/  
int f(int x)  
{int y;  
  y=x;  
  x=x+1;  
  /*@ assert x == \at(x, Pre)+1; */  
  return(y);  
}
```

Listing 31 – contrat

```
/*@ requires x >= 0;  
@ assigns \nothing;  
@ ensures \result == \old(x);  
@*/  
int f(int x)  
{int y;  
  y=x;  
  x=x+1;  
  return(y);  
}
```

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- Playing with variables
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Listing 32 – an1.c

```
//@ assert l1:  $P(x)$ ;  
  x = e(x);  
//@ assert l2:  $Q(x)$ ;
```


Listing 33 – an1.c

```
//@ assert l1: P(x);  
  x = e(x);  
//@ assert l2: Q(x);
```

► $P(x) \Rightarrow WP(x := e(x))(Q(x))$

Listing 34 – an1.c

```
//@ assert l1:  $P(x)$ ;  
  x = e(x);  
//@ assert l2:  $Q(x)$ ;
```

- $P(x) \Rightarrow WP(x := e(x))(Q(x))$
- $P(x) \Rightarrow Q[x \mapsto e(x)]$

Listing 35 – an1.c

```
//@ assert l1: P(x);  
  x = e(x);  
//@ assert l2: Q(x);
```

- ▶ $P(x) \Rightarrow WP(x := e(x))(Q(x))$
- ▶ $P(x) \Rightarrow Q[x \mapsto e(x)]$
- ▶ $P(x1) \Rightarrow Q[x \mapsto e(x1)]$ (renaming of free occurrences of x by $x1$)

Listing 36 – an1.c

```
//@ assert l1: P(x);  
  x = e(x);  
//@ assert l2: Q(x);
```

- ▶ $P(x) \Rightarrow WP(x := e(x))(Q(x))$
- ▶ $P(x) \Rightarrow Q[x \mapsto e(x)]$
- ▶ $P(x1) \Rightarrow Q[x \mapsto e(x1)]$ (renaming of free occurrences of x by $x1$)
- ▶ $P(x1) \wedge x = e(x1) \Rightarrow Q(x)$

Listing 37 – an1.c

```
//@ assert l1: P(x);  
  x = e(x);  
//@ assert l2: Q(x);
```

- ▶ $P(x) \Rightarrow WP(x := e(x))(Q(x))$
- ▶ $P(x) \Rightarrow Q[x \mapsto e(x)]$
- ▶ $P(x1) \Rightarrow Q[x \mapsto e(x1)]$ (renaming of free occurrences of x by $x1$)
- ▶ $P(x1) \wedge x = e(x1) \Rightarrow Q(x)$
- ▶ $P(x1) \wedge x = e(x1) \Rightarrow Q(x)$

Listing 38 – an1.c

```
//@ assert l1: P(x);  
  x = e(x);  
//@ assert l2: Q(x);
```

- ▶ $P(x) \Rightarrow WP(x := e(x))(Q(x))$
- ▶ $P(x) \Rightarrow Q[x \mapsto e(x)]$
- ▶ $P(x1) \Rightarrow Q[x \mapsto e(x1)]$ (renaming of free occurrences of x by $x1$)
- ▶ $P(x1) \wedge x = e(x1) \Rightarrow Q(x)$
- ▶ $P(x1) \wedge x = e(x1) \Rightarrow Q(x)$
- ▶ $\vdash P(x1) \wedge x = e(x1) \Rightarrow Q(x)$

Listing 39 – an1.c

```
//@ assert l1: P(x);  
  x = e(x);  
//@ assert l2: Q(x);
```

- ▶ $P(x) \Rightarrow WP(x := e(x))(Q(x))$
- ▶ $P(x) \Rightarrow Q[x \mapsto e(x)]$
- ▶ $P(x1) \Rightarrow Q[x \mapsto e(x1)]$ (renaming of free occurrences of x by $x1$)
- ▶ $P(x1) \wedge x = e(x1) \Rightarrow Q(x)$
- ▶ $P(x1) \wedge x = e(x1) \Rightarrow Q(x)$
- ▶ $\vdash P(x1) \wedge x = e(x1) \Rightarrow Q(x)$
- ▶ $P(x1) \wedge x = e(x1) \vdash Q(x)$

Listing 40 – an1.c

```
//@ assert l1: P(x);  
  x = e(x);  
//@ assert l2: Q(x);
```

- ▶ $P(x) \Rightarrow WP(x := e(x))(Q(x))$
- ▶ $P(x) \Rightarrow Q[x \mapsto e(x)]$
- ▶ $P(x1) \Rightarrow Q[x \mapsto e(x1)]$ (renaming of free occurrences of x by $x1$)
- ▶ $P(x1) \wedge x = e(x1) \Rightarrow Q(x)$
- ▶ $P(x1) \wedge x = e(x1) \Rightarrow Q(x)$
- ▶ $\vdash P(x1) \wedge x = e(x1) \Rightarrow Q(x)$
- ▶ $P(x1) \wedge x = e(x1) \vdash Q(x)$
 - Assume {
 $P(x1)$
 - ▶ $x = e(x1)$
 - }
 - Prove: $Q(x)$

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Listing 41 – an1.c

```
int main(void){  
    signed long int x,y,z; //      int x,y,z;  
    x = 1;  
    /*@ assert x == 1; */  
    y = 2;  
    /*@ assert x == 1 && y == 2; */  
    z = x * y;  
    /*@ assert x == 1 && y == 1 && z == 2; */  
    return 0;  
}
```

```
[kernel] Parsing an1.c (with preprocessing)
[wp] Running WP plugin...
[wp] Warning: Missing RTE guards
[wp] 2 goals scheduled
[wp] Proved goals:      4 / 4
    Terminating:      1
    Unreachable:       1
    Qed:                2
```

Goal Assertion 'l1' (file an1.c, line 3):

```
Assume {  
  Type: is_sint32(x) /\ is_sint32(y).  
  (* Initializer *)  
  Init: x = 12.  
  (* Initializer *)  
  Init: y = 24.  
}
```

Prove: $(2 * x) = y$.

Prover Qed returns Valid

Goal Assertion 'l2' (file an1.c, line 5):

```
Assume {
  Type: is_sint32(x) /\ is_sint32(x_1) /\ is_sint32(y).
  (* Initializer *)
  Init: x_1 = 12.
  (* Initializer *)
  Init: y = 24.
  (* Assertion 'l1' *)
  Have: (2 * x_1) = y.
  Have: (1 + x_1) = x.
}
Prove: (2 + y) = (2 * x).
Prover Qed returns Valid
```

```
[kernel] Parsing an1.c (with preprocessing)
[wp] Running WP plugin...
[wp] Warning: Missing RTE guards
[wp] 2 goals scheduled
[wp] Proved goals:      4 / 4
    Terminating:      1
    Unreachable:        1
    Qed:                 2
```

Listing 42 – an2.c

```
void ex(void) {  
    int x=12,y=24;  
    //@ assert l1:  $2*x == y$ ;  
    x = x+1;  
    //@ assert l2:  $y == 2*(x-1)$ ;  
    x = x+2;  
    //@ assert l3:  $y+6 == 2*x$ ;  
}
```

```
[kernel] Parsing an2.c (with preprocessing)
[wp] Running WP plugin...
[wp] Warning: Missing RTE guards
[wp] 3 goals scheduled
[wp] Proved goals:      5 / 5
    Terminating:      1
    Unreachable:       1
    Qed:                3
```


Goal Assertion 'l1' (file an2.c, line 3):

```
Assume {  
  Type: is_sint32(x) /\ is_sint32(y).  
  (* Initializer *)  
  Init: x = 12.  
  (* Initializer *)  
  Init: y = 24.  
}  
Prove: (2 * x) = y.  
Prover Qed returns Valid
```

Goal Assertion 'l2' (file an2.c, line 5):

```
Assume {  
  Type: is_sint32(x) /\ is_sint32(x_1) /\ is_sint32(y).  
  (* Initializer *)  
  Init: x_1 = 12.  
  (* Initializer *)  
  Init: y = 24.  
  (* Assertion 'l1' *)  
  Have: (2 * x_1) = y.  
  Have: (1 + x_1) = x.  
}  
Prove: (2 + y) = (2 * x).  
Prover Qed returns Valid
```

Goal Assertion 'l3' (file an2.c, line 7):

Assume {

Type: $\text{is_sint32}(x) \wedge \text{is_sint32}(x_1) \wedge \text{is_sint32}(x_2) \wedge \text{is_s}$

(* Initializer *)

Init: $x_2 = 12$.

(* Initializer *)

Init: $y = 24$.

(* Assertion 'l1' *)

Have: $(2 * x_2) = y$.

Have: $(1 + x_2) = x_1$.

(* Assertion 'l2' *)

Have: $(2 + y) = (2 * x_1)$.

Have: $(2 + x_1) = x$.

}

Prove: $(6 + y) = (2 * x)$.

Prover Qed returns Valid

Listing 43 – an2bis.c

```
void ex(void) {  
    int x=12,y=24;  
    //@ assert l1:  $2*x == y$ ;  
    x = x+1;  
    //@ assert l2:  $y == 2*(x-1)$ ;  
    x = x+2;  
    //@ assert l3:  $y+6 == 2*x$ ;  
}
```

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- ▶ Hoare Model : `-wp` hoare is the option of frama-c
- ▶ Typed Model : default model is typed model

- ▶ It simply maps each C variable to one pure logical variable.
- ▶ Heap cannot be represented in this model, and expressions such as `*p` cannot be translated at all.
- ▶ You can still represent pointer values, but you cannot read or write the heap through pointers.

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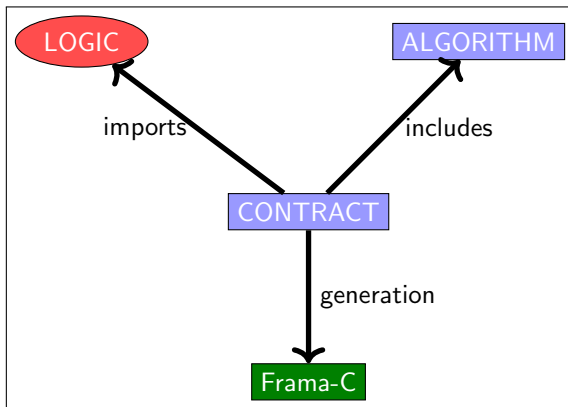
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▶ predicate

(Predicate)

Listing 44 – project-divers/predicate1.c

```
/*@ predicate is_positive(integer x) = x > 0; */  
  
/*@ logic integer get_sign(real x) = @ x > 0.0 ? 1 : (x < 0.0 ? -1 : 0);  
*/  
/*@ logic integer max(int x, int y) = x > y ? x : y;  
*/
```

(Lemma)

Listing 45 – project-divers/lemma1.c

```
/*@ lemma div_mul_identity:  
@ \forall real x, real y; y != 0.0 => y*(x/y) == x; @*/  
  
/*@ lemma div_qr:  
@ \forall int a, int b; a >= 0 && b > 0 =>  
@ \exists int q, int r; a == b*q + r && 0 <= r && r < b; @*/
```

(Definition of fibonacci function)

Listing 46 – project-divers/predicate2.c

```
/*@ axiomatic mathfibonacci{  
  @ logic integer mathfib(integer n);  
  @ axiom mathfib0: mathfib(0) == 1;  
  @ axiom mathfib1: mathfib(1) == 1;  
  @ axiom mathfibrec: \forall integer n; n > 1  
  ==> mathfib(n) == mathfib(n-1)+mathfib(n-2);  
  @ } */
```

(Definition of gcd)

Listing 47 – project-divers/predicate3.c

```
/*@ inductive is_gcd(integer a, integer b, integer d) {  
  @ case gcd_zero:  
  @ \forall integer n; is_gcd(n,0,n);  
  @ case gcd_succ:  
  @ \forall integer a,b,d; is_gcd(b, a % b, d) ==> is_gcd(a,b,d); @}  
  @ */
```

(Definition of function odd/even)

Listing 48 – project-divers/predicate4.c

```
//@ predicate pair(integer x) = (x/2)*2==x;
//@ predicate impair(integer x) = (x/2)*2!=x;
//@ lemma ex: \forall integer a,b; a < b ==> 2*a < 2*b;

/*@ inductive is_gcd(integer a, integer b, integer c) {
  case zero: \forall integer n; is_gcd(n,0,n);
  case un: \forall integer u,v,w; u >= v ==> is_gcd(u-v,v,w);
  case deux: \forall integer u,v,w; u < v ==> is_gcd(u,v-u,w);
}
*/
```

STATES l'ensemble des états sur l'ensemble X des variables, S une instruction de programme sur X , Q une propriété d'états, $s \in STATES$.

$$wp(S)(Q)(s) \stackrel{def}{=} \left(\begin{array}{l} \forall t \in STATES : \mathcal{D}(S)(s) = t \Rightarrow tQ(t) \\ \wedge \\ \exists t \in STATES : \mathcal{D}(S)(s) = t \end{array} \right) \dots$$

$$\begin{aligned} wp(S)(Q)(s) &\stackrel{def}{=} \left(\begin{array}{l} \forall t \in STATES : \mathcal{D}(S)(s) = t \Rightarrow tQ(t) \\ \wedge \\ \exists t \in STATES : \mathcal{D}(S)(s) = t \end{array} \right) \dots \dots \\ wlp(S)(Q)(s) &\stackrel{def}{=} (\forall t \in STATES : \mathcal{D}(S)(s) = t \Rightarrow Q(t)) \\ \dots \dots \dots \\ wlp(S)(Q)(s) &\equiv \overline{(\exists t \in STATES : \mathcal{D}(S)(s) = t \wedge \overline{Q}(t))} \dots \dots \end{aligned}$$

STATES l'ensemble des états sur l'ensemble X des variables, S une instruction de programme sur X, Q une propriété d'états, $s \in STATES$.

$$wp(S)(Q)(s) \stackrel{def}{=} \left(\begin{array}{l} \forall t \in STATES : \mathcal{D}(S)(s) = t \Rightarrow tQ(t) \\ \wedge \\ \exists t \in STATES : \mathcal{D}(S)(s) = t \end{array} \right) \dots \dots$$

$$wlp(S)(Q)(s) \stackrel{def}{=} (\forall t \in STATES : \mathcal{D}(S)(s) = t \Rightarrow Q(t))$$

$$\dots \dots \dots$$

$$wlp(S)(Q)(s) \equiv \overline{(\exists t \in STATES : \mathcal{D}(S)(s) = t \wedge \overline{Q}(t))} \dots \dots \dots$$

- ▶ $loop(S) \equiv \overline{(\exists t \in STATES : \mathcal{D}(S)(s) = t)}$ (ensemble des états qui ne permettent pas à S de terminer)
 - ▶ $wp(S)(Q)(s) \equiv wlp(S)(Q)(s) \wedge \overline{loop(S)}(s)$
-

Weakest Liberal Precondition of S for P

$$\{S\}P(x) \stackrel{def}{=} \forall x_f. x \xrightarrow{S} x_f \Rightarrow P(x_f)$$

.....
☒ Definition triplets de Hoare Correction Totale

$$[P]\mathbf{S}[Q] \stackrel{def}{=} P \Rightarrow wp(S)(Q)$$

.....

Correction

:

Si $[P]\mathbf{S}[Q]$ est dérivé selon les règles ci-dessus, alors $P \Rightarrow wp(S)Q$.

- ▶ $[P(e/x)]\mathbf{X} := \mathbf{E}(\mathbf{X})[P]$ est valide : $wp(X := E)(P)/x = P(e/x)$.
- ▶ $[\exists n \in \mathbb{N}. P(n)]\mathbf{while\ B\ do\ S\ od}[P(0)]$: si s est un état de $P(n)$ alors au bout de n boucles on atteint un état s_f tel que $P(0)$ est vrai en s_f .

Complétude

:

Si $P \Rightarrow wp(S)(Q)$, alors il existe une preuve de $[P]\mathbf{S}[Q]$ construites avec les règles ci-dessus,

- ▶ $P \Rightarrow wp(X := E(X))(Q) : P \Rightarrow Q(e/x)$ et $[Q(e/x)]\mathbf{X} := \mathbf{E}(\mathbf{X})[Q]$ constituent une preuve.
- ▶ $P \Rightarrow wp(\text{while})(Q) :$
 - On construit la suite de $P(n)$ en définissant $P(n) = W_n$.
 - On vérifie que cela vérifie la règle du while.

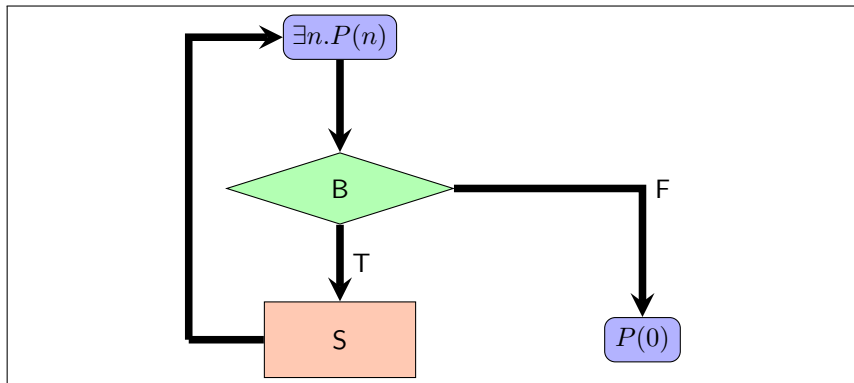
Introducing a variant

Introducing a variant

- ▶ assignment and skip are terminating
- ▶ If S_1 and S_2 are terminating, then $S_1 ; S_2$ is terminating and $\text{if}(B, S_1, S_2)$ is terminating

Introducing a variant

- ▶ assignment and skip are terminating
- ▶ If $S1$ and $S2$ are terminating, then $S1 ; S2$ is terminating and $\text{if}(B, S1, S2)$ is terminating



- ▶ An assertion Q over $\mathbb{S}$ maps states of $\mathbb{S}$ to booleans of $\mathbb{B}$:
 $Q \in \mathbb{S} \longrightarrow \mathbb{B}$
- ▶ $\mathbb{A} \stackrel{def}{=} \mathbb{S} \longrightarrow \mathbb{B}$
- ▶ $\Rightarrow$ defines a partial ordering over $\mathbb{A}$
- ▶ $(\mathbb{A}, \Rightarrow)$ is a complete lattice with a smallest element $\perp$, a greatest element $\top$.
- ▶ For any Q in $\mathbb{A}$, $set(Q) = \{s \in \mathbb{S} | Q(s) = 1\}$ and $(\mathcal{P}(\mathbb{S}), \subseteq)$ is a complete lattice and $Q1 \Rightarrow Q2 \equiv set(Q1) \subseteq set(Q2)$
- ▶ $w(B)(S) \stackrel{def}{=} \text{while } B \text{ do } S \text{ od}$
- ▶ $F_{B,S}(X) = B \wedge wp(S)(X) \vee \neg B \wedge Q$
- ▶ $F_{B,S}(\bigcup_{i \in I} X_i) = \bigcup_{i \in I} F_{B,S}(X_i)$ for any index sequence of assertions $A_i \ i \in I$.
- ▶ $F_{B,S}$ is a continuous map and has a least fixed-point by Kleene's theorem.
- ▶ $lfp(F_{B,S}) = \bigcup_{i \in \mathbb{N}} F_{B,S}^i$ with $F_{B,S}^0 = \perp$ and $F_{B,S}^{i+1} = F_{B,S}(F_{B,S}^i)$ for any $i \in \mathbb{N}$.

Defining approximations of while loop

- ▶ $F_{B,S}^0 = \perp$: the first approximation means that the loop is diverging
- ▶ $F_{B,S}^{i+1} = B \wedge wp(S)(F_{B,S}^i) \vee \neg B \wedge Q$
- ▶ $P(0) \stackrel{def}{=} F_{B,S}^1 \equiv \neg B \wedge Q$
- ▶ ...
- ▶ $P(i+1) \stackrel{def}{=} B \wedge wp(S)(P(i))$

The sequence of assertions $P(i)$ satisfies the following properties

- ▶ $\forall i \in \mathbb{N}, [P(i+1)]S[P(i)]$
- ▶ $\forall i \in \mathbb{N}, [P(i+1)] \rightarrow B$
- ▶ $P(0) \Rightarrow \neg B$

By the rule of Hoare Logic for Total Correctness, we get the following property $[\exists i \in \mathbb{N}, P(i)]w(B)(S)[P(0)]$.

- ▶ The termination is proved by showing that each loop terminates.
- ▶ Any loop is characterized by an expression `expvariant(x)` called `variant` which should decrease each execution of the body :

$$\forall x_1, x_2. b(x_1) \wedge x_1 \xrightarrow{S} x_2 \Rightarrow \text{expvariant}(x_1) > \text{expvariant}(x_2)$$

(Variant)

Listing 49 – project-divers/variant1.c

```
/*@ requires n > 0;
    terminates n > 0;

    ensures \result == 0;
*/
int code(int n) {
    int x = n;
    /*@ loop invariant x >= 0 && x <= n;
        loop assigns x;
        loop variant x;
    */
    while (x != 0) {
        x = x - 1;
    };
    return x;
}
```

(Variant)

Listing 50 – project-divers/variant3.c

```
int f() {  
  int x = 0;  
  int y = 10;  
  /*@  
    loop invariant  
    0 <= x < 11 && x+y == 10;  
    loop variant y;  
  */  
  while (y > 0) {  
    x++;  
    y—;  
  }  
  return 0;  
}
```

(The ANSI/ISO C Specification Language (ACSL)) (30 avril 2026) (Dominique Méry)

(30 avril 2026) (Dominique Méry)

MALG & MOVEX

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- ▶ Defining the mathematical function to compute *mathf*
- ▶ Stating the postcondition using the mathematical function
- ▶ Evaluating the inductive sequence u_i computing the function *mathf*
- ▶ $\forall i \in \mathbb{N} : u_i = \text{mathf}(i)$
- ▶ Evaluating relationship among variables.

(power2.h)

Listing 53 – project-powers/power21.h

```
#ifndef _A.H
#define _A.H
// Definition of the mathematical function mathpower2
/*@ axiomatic mathpower2 {
  @ logic integer mathpower2(integer n);
  @ axiom mathpower2_0: mathpower2(0) == 0;
  @ axiom mathpower2_rec: \forall integer n; n > 0
  ==> mathpower2(n) == mathpower2(n-1) + n+n+1;
  @ } */
/*@ axiomatic matheven {
  @ logic integer matheven(integer n);
  @ axiom matheven_0: matheven(0) == 0;
  @ axiom matheven_rec: \forall integer n; n > 0
  ==> matheven(n) == matheven(n-1) + 2;
  @ } */
// We define v and w in a one shot axiomatic definition
/*@ axiomatic vw {
  @ logic integer v(integer n);
  @ logic integer w(integer n);
  @ axiom v_0: v(0) == 0;
  @ axiom w_0: w(0) == 0;
  @ axiom v_rec: \forall integer n; n > 0
  ==> v(n) == v(n-1) + n+n+1 && w(n) == w(n-1) + 2;
  @ } */
```

(power2.h)

Listing 54 – project-powers/power22.h

```
/*@ lemma propw:
@ \forall int n; n >= 0 => w(n) == n+n; @*/
/*@ lemma propv:
@ \forall int n; n >= 0 => v(n) == n*n; @*/
/*@ lemma prop1:
@ \forall int n; n >= 0 => matheven(n) == n+n; @*/
/*@ lemma prop2:
@ \forall int n; n >= 0 => mathpower2(n) == n*n; @*/
/*@ axiomatic auxmath {
  @ lemma rule1: \forall int n; n > 0 => n*n == (n-1)*(n-1)+2*(n-1)+1;
  @ } */
/*@ requires 0 <= x;
   assigns \nothing;
   ensures \result == x*x;
*/
int power2(int x);
#endif
```

(power2.h)

Listing 55 – project-powers/power2.c

```
#include <limits.h>
#include "power2.h"

int power2(int x)
{
    int r, k, cv, cw, or, ok, ocv, ocw;
    r=0; k=0; cv=0; cw=0; or=0; ok=k; ocv=cv; ocw=cw;
    /*@ loop invariant 0 <= cv && 0 <= cw && 0 <= k;
       @ loop invariant cv == k*k;
       @ loop invariant k <= x;
       @ loop invariant cw == 2*k;
       @ loop invariant 4*cv == cw*cw;
       @ loop assigns k, cv, cw, or, ok, ocv, ocw;
       @ loop variant x-k;

    */
    while (k<x)
    {
        ok=k; ocv=cv; ocw=cw;
        k=ok+1;
        cv=ocv+ocw+1;
        cw=ocw+2;

    }
    r=cv;
    return (r);
}
```

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⑧ Conclusion

- (The ANSI/ISO C Specification Language (ACSL)) (30 avril 2026) (Dominique Méry)

- ▶ Defining domain properties (axioms, lemmas, proofs)
- ▶ Defining loop invariants (typing, equation, ...)
- ▶ Analyzing inductive properties
- ▶ Identifying inputs (*requires*) and outputs (*ensures*)